

$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n}): \text{resonances}$ [1975Sc40](#), [1971Mc23](#), [1964Ku12](#) $J^\pi = 1/2^+$ ^{31}P ground state.

[1975Sc40](#): $^{31}\text{P}(\alpha, \text{p})$ with 3.25-5.55-MeV α beams produced from the 5.5-MV Van de Graaff accelerator of the University of Lowell. The target was prepared by evaporating natural ^{31}P onto carbon backings. Protons were detected using four silicon surface-barrier detectors. Measured proton yields and angular distributions. Deduced resonance levels at $E_x = 9.9$ -11.8 MeV, J, and π .

[1971Mc23](#): $^{31}\text{P}(\alpha, \text{p})$ with 3.25-5.25-MeV α beams produced from the 5.5-MeV Van de Graaff accelerator of the Southern Universities Nuclear Institute. The target was Zn_3P_2 evaporated onto carbon backings. Protons were detected using four solid state detectors. Measured proton yields and angular distributions. Deduced resonance levels at $E_x = 9.9$ -11.7 MeV, J, and resonance strengths.

[1964Ku12](#): $^{31}\text{P}(\alpha, \text{p})$ with 1.7-3.3-MeV α beams produced from the Utrecht Van de Graaff accelerator. A zinc phosphide target was prepared by evaporation of 40-500- mg/cm^2 Zn_3P_2 onto solid copper backings. Protons were detected using five silicon surface barrier counters at 87° , 120° , 135° , 150° , and 172° . Measured resonance yields and proton angular distributions. Deduced ^{35}Cl resonance energies, J, and resonance strengths.

[1970Um01](#): $^{34}\text{S}(\text{p}, \text{n})$, $^{31}\text{P}(\alpha, \text{n})$, $^{31}\text{P}(\alpha, \alpha_0)$ reactions with proton and α beams produced from the Florida State University Tandem Van de Graaff accelerator. Targets for $^{34}\text{S}(\text{p}, \text{n})$ were a CdS target by evaporating ^{34}S enriched CdS onto 0.12-mm gold plate and a AgS target heating ^{34}S (67% enriched) in the presence of 0.025-mm Ag foil. Targets for $^{31}\text{P}(\alpha, \text{n})$ were prepared by evaporating natural Ca_3P_2 onto 0.075-mm Ta plate and 0.025-mm Al foil. Positrons from ^{34}Cl decay were detected using an NE102 plastic scintillator. γ rays in ^{34}S following the decay were detected using a Ge(Li) detector. Measured excitation functions. Deduced level energies and isospin $T=3/2$ for ^{35}Cl resonance states at 12.9 and 13.9 MeV.

 ^{35}Cl Levels

$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma$: from [1964Ku12](#) for $E_x < 9950$ and from [1971Mc23](#) for $E_x > 9950$.

E(level) [†]	J^π [‡]	$E_\alpha(\text{lab})$ (keV) ^{&}	Comments
9127 9	$5/2^\#$	2404 [#] 10	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 0.16$ eV.
9161 4	$1/2^\#$	2442 [#] 4	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 1.6$ eV.
9256 4	$1/2^\#$	2550 [#] 4	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 2.6$ eV.
9400 9	$1/2^\#$	2712 [#] 10	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 0.6$ eV.
9456 4	$3/2^\#$	2776 [#] 4	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 2.8$ eV.
9481.0 18	$3/2^\#$	2804 [#] 2	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 18$ eV.
9551 6	$5/2^\#$	2883 [#] 7	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 1.9$ eV.
9673 6	$1/2^\#$	3021 [#] 7	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 1.1$ eV.
9713.0 27	$1/2^\#$	3066 [#] 3	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 41$ eV.
9751.1 27	$7/2^\#$	3109 [#] 3	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 7.6$ eV.
9814.0 27	$5/2^\#$	3180 [#] 3	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 20$ eV.
9869.8 27	$1/2$	3243 3	$E_\alpha(\text{lab})$ (keV): weighted average of 3243 3 (1964Ku12) and 3242 5 (1975Sc40). J^π : $1/2$ from 1964Ku12 . $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 46$ eV.
9900.8 27	$3/2$	3278 3	$E_\alpha(\text{lab})$ (keV): weighted average of 3278 3 (1964Ku12) and 3278 5 (1975Sc40). J^π : $3/2$ from 1964Ku12 . $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 38$ eV.
9923 4	$3/2^-$	3303 4	$E_\alpha(\text{lab})$ (keV): weighted average of 3302 4 (1964Ku12) and 3305 5 (1975Sc40). J^π : $(3/2)$ from 1964Ku12 . $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 80$ eV.
9958 6	$3/2$	3342 7	$E_\alpha(\text{lab})$ (keV): weighted average of 3348 5 (1971Mc23) and 3335 5 (1975Sc40). J^π : $3/2$ from 1971Mc23 . $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 34$ eV.
9968 5	$(1/2, 3/2)^\oplus$	3354 5	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma = 17$ eV.
10031 5	$1/2^+, 3/2^-$	3425 5	$E_\alpha(\text{lab})$ (keV): weighted average of 3424 5 (1971Mc23) and 3425 5 (1975Sc40). J^π : $(1/2)$ from 1971Mc23 .

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$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances **1975Sc40, 1971Mc23, 1964Ku12** (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _α (lab) (keV) ^{&}	Comments
			(2J+1)Γ _α Γ _p /Γ=124 eV.
10058 5	3/2 [@]	3455 5	(2J+1)Γ _α Γ _p /Γ=40 eV.
10075 5	(1/2) [@]	3475 5	(2J+1)Γ _α Γ _p /Γ=52 eV.
10090 5	5/2 ⁺	3491 5	E _α (lab) (keV): weighted average of 3491 5 (1971Mc23) and 3490 5 (1975Sc40). J ^π : 5/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=63 eV.
10132 5	3/2 ⁻	3539 5	E _α (lab) (keV): weighted average of 3538 5 (1971Mc23) and 3540 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=165 eV.
10172 5	3/2 ⁻ , 5/2 ⁻	3584 5	E _α (lab) (keV): weighted average of 3587 5 (1971Mc23) and 3580 5 (1975Sc40). J ^π : (5/2) from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=26 eV.
10215 5	3/2	3633 5	E _α (lab) (keV): weighted average of 3631 5 (1971Mc23) and 3635 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=63 eV.
10236 5	3/2 ⁻	3656 5	E _α (lab) (keV): weighted average of 3656 5 (1971Mc23) and 3655 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=271 eV.
10278 5	1/2 [@]	3704 5	(2J+1)Γ _α Γ _p /Γ=15 eV.
10295 5		3723 5	
10322 5	5/2 ⁺ , 7/2 ⁻	3753 5	E _α (lab) (keV): weighted average of 3755 5 (1971Mc23) and 3750 5 (1975Sc40). J ^π : 5/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=63 eV.
10340 5	(3/2, 5/2) [@]	3774 5	(2J+1)Γ _α Γ _p /Γ=50 eV.
10359 5	3/2	3795 5	E _α (lab) (keV): weighted average of 3795 5 (1971Mc23) and 3795 5 (1975Sc40). J ^π : (3/2) from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=155 eV.
10385 5		3825 5	
10399 5	3/2 ⁺ , 7/2 ⁺	3841 5	E _α (lab) (keV): weighted average of 3842 5 (1971Mc23) and 3840 5 (1975Sc40). J ^π : (1/2, 3/2) from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=61 eV.
10431 5	5/2 ⁺	3877 5	E _α (lab) (keV): weighted average of 3876 5 (1971Mc23) and 3877 5 (1975Sc40). J ^π : 5/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=66 eV.
10467 5	5/2 ⁺	3917 5	E _α (lab) (keV): weighted average of 3918 5 (1971Mc23) and 3915 5 (1975Sc40). J ^π : 5/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=211 eV.
10486 5	5/2 ⁺	3939 5	E _α (lab) (keV): weighted average of 3939 5 (1971Mc23) and 3938 5 (1975Sc40). J ^π : 5/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=76 eV.
10534 5	1/2 ⁺ , 5/2 ⁺	3993 5	E _α (lab) (keV): weighted average of 3990 5 (1971Mc23) and 3995 5 (1975Sc40). J ^π : 1/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=182 eV.
10587 5	3/2 ⁺	4053 5	E _α (lab) (keV): weighted average of 4050 5 (1971Mc23) and 4055 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=71 eV.
10641 5	5/2 ⁺	4114 5	E _α (lab) (keV): weighted average of 4113 5 (1971Mc23) and 4114 5 (1975Sc40). J ^π : 5/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=272 eV.
10678 5	3/2 ⁺ , 5/2 ⁺	4156 5	E _α (lab) (keV): weighted average of 4156 5 (1971Mc23) and 4155 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 .
			(2J+1)Γ _α Γ _p /Γ=132 eV.
10696 5	7/2 ⁻	4176 5	E _α (lab) (keV): weighted average of 4174 5 (1971Mc23) and 4178 5 (1975Sc40).

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$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances **1975Sc40, 1971Mc23, 1964Ku12** (continued) ^{35}Cl Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$E_\alpha(\text{lab}) \text{ (keV)}^\&$	Comments
10734 5	$3/2^+, 9/2^+$	4219 5	J^π : 7/2 from 1971Mc23. (2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=100$ eV. $E_\alpha(\text{lab})$ (keV): weighted average of 4218 5 (1971Mc23) and 4220 5 (1975Sc40).
10760 5	$3/2^+, 7/2^-$	4248 5	J^π : 3/2 from 1971Mc23. (2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=269$ eV. $E_\alpha(\text{lab})$ (keV): weighted average of 4246 5 (1971Mc23) and 4250 5 (1975Sc40).
10800 5	$5/2^+$	4293 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4290 5 (1971Mc23) and 4295 5 (1975Sc40).
10816 5	$1/2^+, 3/2^-$	4311 5	J^π : 5/2 from 1971Mc23. (2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=238$ eV. $E_\alpha(\text{lab})$ (keV): weighted average of 4308 5 (1971Mc23) and 4313 5 (1975Sc40).
10844 5	$3/2^-$	4343 5	J^π : 3/2 from 1971Mc23. (2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=345$ eV. $E_\alpha(\text{lab})$ (keV): weighted average of 4342 5 (1971Mc23) and 4343 5 (1975Sc40).
10870 5	$(5/2, 7/2)^\text{@}$	4372 5	J^π : 3/2 from 1971Mc23. (2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=923$ eV.
10886 5		4390 5	(2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=137$ eV.
10909 5	$3/2^+, 5/2^-$	4416 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4413 5 (1971Mc23) and 4418 5 (1975Sc40).
10928 5	$7/2^\text{@}$	4438 5	(2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=231$ eV.
10942 5		4454 5	
10964 5	$5/2^+$	4478 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4478 5 (1971Mc23) and 4477 5 (1975Sc40).
10973 5		4489 5	J^π : 5/2 from 1971Mc23. (2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=181$ eV.
10998 5	$3/2^+, 7/2^-$	4517 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4517 5 (1971Mc23) and 4517 5 (1975Sc40).
11034 5	$3/2^+$	4557 5	J^π : (5/2) from 1971Mc23. (2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=172$ eV. $E_\alpha(\text{lab})$ (keV): weighted average of 4556 5 (1971Mc23) and 4558 5 (1975Sc40).
11063 5	$1/2^+, 9/2^+$	4590 5	J^π : 3/2 from 1971Mc23. (2J+1) $\Gamma_\alpha\Gamma_p/\Gamma=218$ eV.
11082 5	$5/2^+, 9/2^+$	4612 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4589 5 (1971Mc23) and 4590 5 (1975Sc40).
11105 5		4638 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4611 5 (1971Mc23) and 4613 5 (1975Sc40).
11123 5	$3/2$	4658 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4657 5 (1971Mc23) and 4658 5 (1975Sc40).
11142 5	$5/2^+$	4679 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4678 5 (1971Mc23) and 4680 5 (1975Sc40).
11154 5		4693 5	
11169 5		4710 5	
11179 5	$3/2^-, 5/2^+$	4721 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4721 5 (1971Mc23) and 4720 5 (1975Sc40).
11185 5		4728 5	
11198 5	$3/2$	4743 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4745 5 (1971Mc23) and 4740 5 (1975Sc40).
11230 5		4779 5	
11246 5	$3/2$	4797 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4799 5 (1971Mc23) and 4795 5 (1975Sc40).
11265 5	$3/2^-$	4818 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4820 5 (1971Mc23) and 4815 5 (1975Sc40).
11287 5		4843 5	
11307 5		4866 5	
11314 5	$1/2^-, 5/2^+$	4874 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4875 5 (1971Mc23) and 4873 5 (1975Sc40).
11327 5	$1/2^-, 5/2^+$	4888 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4886 5 (1971Mc23) and 4890 5 (1975Sc40).
11357 5		4922 5	
11375 5		4942 5	
11390 5		4960 5	
11410 5		4982 5	
11421 5	$3/2^-$	4995 5	$E_\alpha(\text{lab})$ (keV): weighted average of 4995 5 (1971Mc23) and 4995 5 (1975Sc40).
11434 5		5009 5	
11460 5		5038 5	
11473 5	$1/2^-, 5/2^-$	5053 5	$E_\alpha(\text{lab})$ (keV): weighted average of 5056 5 (1971Mc23) and 5050 5 (1975Sc40).

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$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances [1975Sc40](#), [1971Mc23](#), [1964Ku12](#) (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _α (lab) (keV) ^{&}	Comments
11492 5		5075 5	
11505 5	3/2 ⁻ , 5/2 ⁻	5089 5	E _α (lab) (keV): weighted average of 5090 5 (1971Mc23) and 5088 5 (1975Sc40).
11525 5		5112 5	
11540 5		5129 5	
11551 5	3/2 ⁻ , 5/2 ⁺	5141 5	E _α (lab) (keV): weighted average of 5142 5 (1971Mc23) and 5140 5 (1975Sc40).
11565 5		5157 5	
11589 5		5184 5	
11609 5	5/2 ⁺ , 7/2 ⁻	5207 5	E _α (lab) (keV): weighted average of 5209 5 (1971Mc23) and 5205 5 (1975Sc40).
11627 5	1/2 ⁺ , 5/2 ⁺	5227 5	E _α (lab) (keV): weighted average of 5224 5 (1971Mc23) and 5230 5 (1975Sc40).
11638 5		5239 5	
11651 5		5254 5	
11684 5	1/2 ⁺ , 5/2 ⁺	5292 ^a 5	
11773 5	3/2 ⁻ , 7/2 ⁻	5392 ^a 5	
11783 5	3/2 ⁻	5403 ^a 5	
12900 ^b			T=3/2
13900 ^b			T=3/2

[†] From E_{c.m.}+S(α)(^{35}Cl), where S(α)(^{35}Cl)=6997.91 4 ([2021Wa16](#)). E_{c.m.}=E_α(lab)×m(^{31}P)/[m_α+m(^{31}P)].

[‡] From [1975Sc40](#) by analyzing measured angular distributions, unless otherwise noted.

From [1964Ku12](#).

@ From [1971Mc23](#).

& From [1971Mc23](#), unless otherwise noted.

^a From [1975Sc40](#).

^b From [1970Um01](#). Two ^{35}Cl resonances are found in $^{34}\text{S}(\text{p}, \text{n})$ and $^{31}\text{P}(\alpha, \text{n})$, but not in $^{31}\text{P}(\alpha, \alpha_0)$ reactions. T=1 ^{34}S g.s. and T=1/2 protons can populate either T=1/2 or T=3/2 states in ^{35}Cl . T=1/2 ^{31}P g.s. and T=0 α particles can populate T=1/2 states in ^{35}Cl . A small isospin impurity would allow formation of T=3/2 states in ^{35}Cl . The lack of resonances in $^{31}\text{P}(\alpha, \alpha_0)$ indicates that these two resonances are most likely T=3/2. A T=3/2 resonances in $^{31}\text{P}(\alpha, \alpha_0)$ scattering is doubly isospin forbidden (both entrance and exit channels) compared with singly forbidden (entrance channel only) for $^{31}\text{P}(\alpha, \text{n})$.